

Article

Analysis of Sewage Sludge Drying Parameters Using Different Additives

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Abstract

This paper describes the process of drying sewage sludge mixtures with the addition of various components: straw chaff, wood sawdust, ash, bark, wood chips, and walnut shells. The tests were conducted in two series: summer and autumn (with maximum insolation of 24.1 and 29.8 MJ·m⁻², respectively). Using a set of sensors with which the experimental station was equipped, the parameters of the environment (temperature, humidity, and insolation) and the parameters of the dried mixtures (temperature and humidity) were measured. Based on the results obtained, the influence of external factors on the parameters, time, and drying effect of the respective mixtures was analyzed. With the initial moisture content of the mixtures ranging from 41 to 79%, a final moisture content of 6 to 49% was obtained, depending on the components and drying conditions. It was found that the drying rate was most influenced by the amount of solar energy and the associated outdoor (maximum 29 °C and 19 °C) and indoor (maximum 33 °C and 24 °C) air temperatures, and in the second series, there was an additional effect of the temperature of the mixtures (maximum 30 °C), upon which the intensity of water evaporation depended. Straw chaff and walnut shells proved to be the best additives, for which the highest drying rates were obtained (max. 50 to 60% humidity drop). The possibility of using dried materials for agricultural and energy purposes was indicated. This approach is in line with the principles of sustainable development.



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Keywords: sewage sludge; solar drying; drying parameters; drying rate

1. Introduction

Cities and, to a lesser extent, rural population centers with wastewater treatment plants are struggling to manage the increasing mass of sludge generated as waste in mechanical and biological wastewater treatment [1,2]. As a rule, sewage sludge contains a large amount of water, and the dry matter is dominated by organic matter over mineral matter, making this material attractive for use as fertilizer in agriculture [3]. However, it should be emphasized that the composition of different sewage sludges depends strongly on the type of wastewater and its treatment technology [4,5]. The peculiar structure of dewatered sludge, which is largely composed of cells of living organisms (mainly bacteria) of activated sludge, means that even after mechanical dewatering, its mass still contains 80% (and often more) water [6,7]. This means that one popular method of sludge management, i.e., incineration, may be inefficient from an energy standpoint [8–10].

While sewage sludge accounts by volume for only 3% of the amount of wastewater flowing into the treatment plant, the pollutant load contained within it exceeds the total load in wastewater by more than a half, which translates into a challenge facing wastewater treatment plants in terms of disposal and management. Among the popular methods of sewage sludge management, it has long been used as a field fertilizer—as much as 27.1% of municipal sludge in 2022 [11]. Since the 1990s, Poland has also seen the development of composting technology [12,13]. A much more advanced technology, but also more expensive to invest in than composting due to the high complexity of the installation, is biogas plants [14,15]. The use of sewage sludge in the fermentation process allows for significant economic and energy benefits through the sale of the electricity and heat produced or, alternatively, the biomethane produced, which can substitute a significant portion of the natural gas consumed in Poland [16,17]. Methane fermentation of sewage sludge (and also the production of biohydrogen through dark fermentation) is also becoming increasingly popular worldwide [18]. However, it appears that the technology with very high potential for implementation is sludge drying [19].

Bennamoun [20] believes that drying is a key process for sludge management due to the fact that it has the effect of significantly reducing the weight and volume of the final product, resulting in lower costs of storage, handling, and transportation. An additional important aspect is that during drying processes, the calorific value of sewage sludge increases (about 9–13 MJ·kg^{−1}, a value similar to brown coal), which makes the product suitable for use as fuel in coal-fired power plants, municipal waste incinerators, or cement plants, among others. The law treats sludge as biomass, which means that it is carbon-neutral fuel [21]. This action is in line with the IEA report [22] and Polish law [23]. Thus, an increasingly attractive source of renewable energy is seen in sediments, or more precisely in the organic matter they contain—this is in fact the case, provided that water is evaporated from them beforehand [24,25]. Water evaporation is a key process for highly hydrated sewage sludge with variable heavy metal content and varying degrees of sanitary risk (the highest for raw sludge, the lowest for stabilized and hygienized sludge) [26–29]. It is also worth mentioning that the costs associated with sewage sludge management, especially in the context of water evaporation in thermal processes, can be very high.

Boguniewicz-Zablocka et al. [30] define the issue of solar drying of sludge as an alternative solution to energy-intensive thermal drying technologies due to the economic aspect of using a renewable energy source such as solar radiation. In an era of dwindling reserves of conventional energy sources, at its beginning, the 21st century was already dubbed the century of the sun. Solar dryers, which precisely use the energy of solar radiation to evaporate water from sewage sludge, fit perfectly into this ecological trend. From the standpoint of both capital expenditures and subsequent operating costs, this is currently one of the cheapest ways to reduce the mass and volume of this product [29].

Drying technology, which is simple by design, produces a stable dried sludge with four times less weight than the input charge and a neutral peat-like odor. The technology was first used in Germany in the mid-1990s. Currently, more than half of all sludge dryers are solar dryers. Solar dryers differ in the type of tumbler installed to turn, grind, and transport the dried material inside the hall, as well as in the ventilation system [31].

In Poland, the sun provides an average of 1000 to 1100 kWh of energy for every square meter of flat surface. It is about equivalent to the heat of burning about 110 L of heating oil. The theoretical energy required to evaporate 1 kg of water is 0.627 kWh at normal pressure. When the sludge is fully dried (up to 90–92% DM), the average thermal energy demand ranges from 0.6 to 1.2 kWh·kg^{−1} of evaporated water. Due to the ever-increasing mass of hydrated sewage sludge generated and the difficulty of using it directly, it is advisable to use thermal treatment to reduce the mass. This will allow the achievement

of a hygienization effect and consequently facilitate further management [32–34]. In their literature review, Vijaya VenkataRaman et al. [35] discuss the use of natural solar radiation energy, the application of which has been practiced since time immemorial to dry food and agricultural crops. This fact is also confirmed in [31], which refers to economic issues in terms of the alternative use of the solar drying method instead of conventional methods.

Li et al. [36], after conducting a study of solar drying of sludge, found that the drying effect is affected by insolation, ambient air temperature, and, in addition, the speed of air flow over the sludge layer.

Sludge drying with the addition of various components, including straw chaff, was carried out by [37]. The mixtures were dried under different weather conditions with varying levels of solar radiation. The author found that the drying time depended on the parameters of the environment (temperature and humidity) and varied at different times of the year, and the intensity of the process, regardless of the conditions, was affected by the thickness of the layer of dried material and the frequency of mixing. Any solar dryer, regardless of the type of mechanical equipment installed within it, is nothing more than an effective combination of a horticultural greenhouse with traditional plots where the sludge is dried at most treatment plants to date [38,39]. If the direct influence of variable atmospheric factors is excluded, the sludge dries naturally, and the product is a dried substance with a grain size of ca. 1 to 2 cm and three to four times less weight and volume. The most important element of solar sewage sludge dryers is an efficient ventilation system that significantly supports water evaporation [40,41]. Unfortunately, an efficient ventilation system reduces the water vapor content in the air and lowers the average internal temperature, which reduces the intensity of water evaporation. The application of an additional component to the sewage sludge (e.g., the addition of a structural material) can improve drying conditions during some periods of year (especially fall and winter) [42,43]. Selecting the appropriate method and process conditions is crucial. As a result, the physical and chemical parameters of the resulting mixture can be improved, which can later be used for energy or agricultural purposes [44–46].

The aim of this study was to conduct a comprehensive analysis of the technological parameters of sewage sludge mixtures with biocomponents during solar drying. The research mixture was tested to clarify the effect of the presence of additives on the solar drying process and to evaluate the efficiency of the process and the quality of the dried sludge. A novelty in the presented research was the use of new additives to the dried sludge which, according to the authors, will improve the structure of the dried mixture, increase the surface area of water evaporation, and generally accelerate drying. Hence, the thesis of the paper assumes that the use of admixtures in the form of additives will improve the drying parameters of solar sewage sludge.

2. Materials and Methods

The research was carried out at the experimental station of the Poznań University of Life Sciences in a 10 m² plastic tunnel, divided into three separate research plots of 1.8 × 0.8 × 0.15 m, with a capacity of 216 dm³ each.

The installation was equipped with a system for measuring and controlling the solar drying parameters (Figure 1). Each test station was provided with an individual set of sensors measuring air humidity (10–100%, ±3% RH), pressure (300–1110 hPa ± 1 hPa), temperature (−40–85, ±1 °C), insolation (0–1280 ± 10 W·m²), and the humidity and temperature of the mixture (0–100% ± <2%; −10–85, ±0.5 °C). External conditions were monitored by an individual set of measuring sensors. Data were continuously recorded in the cloud. To improve air circulation inside the tunnel, two (opposite) windows out of 6 were left open, marked as W1 and W2 (Figure 1). Additionally, an electric fan (30 W) was

installed in window W2, which pushed the air out of the tunnel, thus supporting internal air circulation.

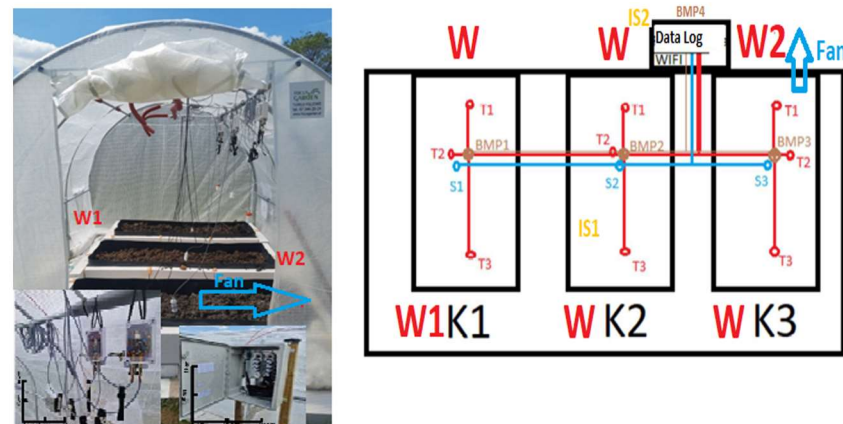


Figure 1. Test tunnel with sludge drying stands. W1, W2—open windows, W—closed window, K—chamber, T—mixture temperature sensor, BMP—humidity, temperature, and air pressure sensor, S—mixture moisture sensor, IS—insolation.

The sewage sludge used in the study was obtained from a municipal wastewater treatment plant. It was sludge that was dewatered on a belt press to a hydration of about 85%. Rye straw chaff, sawdust from coniferous and deciduous trees, landfill ash from power plants, pine bark, beech and alder wood chips, and walnut shells were used as additives to the mixtures. The properties of the materials used in the study are included in Table 1. These materials were selected due to their diverse structures and (with the exception of ash) high organic content.

Table 1. Characteristics of components in dried mixtures.

Mix Component	Mixture Symbol	Moisture [%]	Dry Organic Matter Content [% DM org]	Carbon Content [% DM]	Nitrogen Content [% DM]	Bulk Density, [kg/dm ³]	Dimensions [mm]	Duration of the Series [Day]
Sewage sludge	-	60.74	87.08	37.08	6.57	1.07	-	63
Straw chaff	K1	7.97	92.92	45.98	1.22	0.4	3–50	49
Tree sawdust	K2	7.73	97.83	52.82	1.56	0.15	3–10	49
Ash	K3	7.61	2.44	2.84	0	0.95	0.2–0.5	49
Pine bark	K4	52.05	65.01	50.06	1.13	0.15	10–20	76
Beech–alder wood chips	K5	8.47	73.74	48.19	0.73	0.32	10–16	76
Walnut shells	K6	10.67	70.11	49.89	0.83	0.31	26–39	76
Survey Methodology	-	PN-EN ISO 18134-1 [47]	PN-EN ISO 12880:2004 [48]	PN-EN ISO 21663:2021-06 [49]	PN-EN ISO 21663:2021-06 [49]	PN-EN ISO 1097-3:2000 [50]	PN-EN ISO 933-2:1999 [51]	-

The mixtures were prepared at a volume ratio of sludge to added component of 1:1. The contents of the test quarters were manually mixed daily. The tests were carried out in two series: mixtures K1, K2, and K3 were dried in the first series, while mixtures K4, K5, and K6 were dried in the second series. The first series lasted 48 days during the spring-summer period (May to July), while the second one lasted 80 days during the fall period (September to November). The measurement data of both series were transmitted on average every 10 min or so and recorded using a NodeMCU v3 data logger, which stored the data in the cloud (Thing Speak). In addition to continuous measurement of parameters with sensors,

mixture samples were taken for analysis of moisture content and organic matter content. The analyses in Table 1 were performed in accordance with standards [47–51]. Based on the obtained results of measurements and analysis, drying curves were drawn up for the tested mixtures, the drying rate was calculated, and a correlation analysis of the drying parameters was carried out to evaluate the influence of the respective factors on the process of drying and to compare the drying effects in mixtures with different additives.

3. Results

Table 2 shows the minimum and maximum values of the ambient parameters, as well as the initial and final values of the parameters of the tested mixtures measured during the drying process. The insolation and air temperature values outside the tunnel in series 1 and 2 differed due to the timing of the tests, but the differences between the minimum and maximum temperatures were similar, at 19 and 22 °C, respectively. The temperature extremes inside the tunnel in series 1 and 2 differed by about 10 °C, but the differences between these values for the two series were comparable, at about 25 °C. The humidity of the air inside the tunnel was higher in the first series of tests due to the intensity of evaporation of the mixtures, but it reached a maximum of 100% in both cases.

Table 2. Environment and test mixture parameters.

Serie		SERIE 1			SERIE 2		
Mixture		K1	K2	K3	K4	K5	K6
Insolation, MJ/m ²	Min.		2.57			0.65	
	Max.		24.11			12.9	
Outside temp., °C	Min.		10.67			−3.73	
	Max.		29			19.81	
Inside temp., °C	Min.		8.1			−1.29	
	Max.		33.4			24.2	
Outside humidity, %	Min.		34.25			17.99	
	Max.		99.5			100	
Temp. of mixture, °C	Initial	19.05	18.47	15.7	24.19	24.34	24.05
	Final	25.43	24.38	25.86	3.66	3.81	3.39
	Min.	17.56	18.24	15.7	−0.34	−0.06	−3.46
	Max.	36.12	39.63	32.26	24.19	30.16	24.05
Humidity of mixture, %	Initial	77.27	77.12	41.84	78.23	73.78	78.28
	Final	13.68	12.46	6.23	49.5	28.6	21.93
	Min.	13.68	12.46	5.05	45.63	28.55	20.75
	Max.	79.67	78.99	41.84	80.14	78.16	72.61

Changes in ambient temperature (outside and inside the tunnel) were dependent on insolation and developed according to the season in which the experiment was conducted (Figure 2). In the first series of tests, the external parameters remained at comparable levels, while in the second series, they gradually decreased over time.

An important factor was insolation, which allowed the temperature inside the tunnel to be maintained so that the mixtures could be dried. The temperature of the dried mixtures in series 1 increased, while in series 2 it decreased during the course of the experiment, according to the change in outdoor temperature. By contrast, the moisture content of the

mixtures decreased in both series of tests, reaching lower final values in series 1 than in series 2. The course of changes in moisture content and temperature of the mixtures is shown in Figure 3.

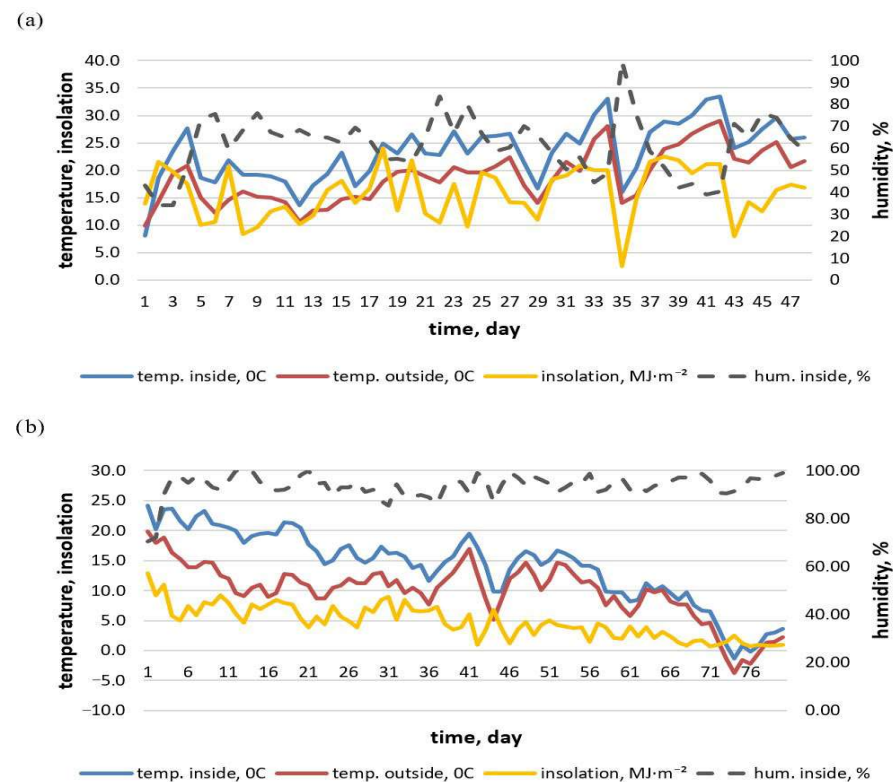


Figure 2. Changes in environment temperatures and insolation in research series 1 (a) and 2 (b).

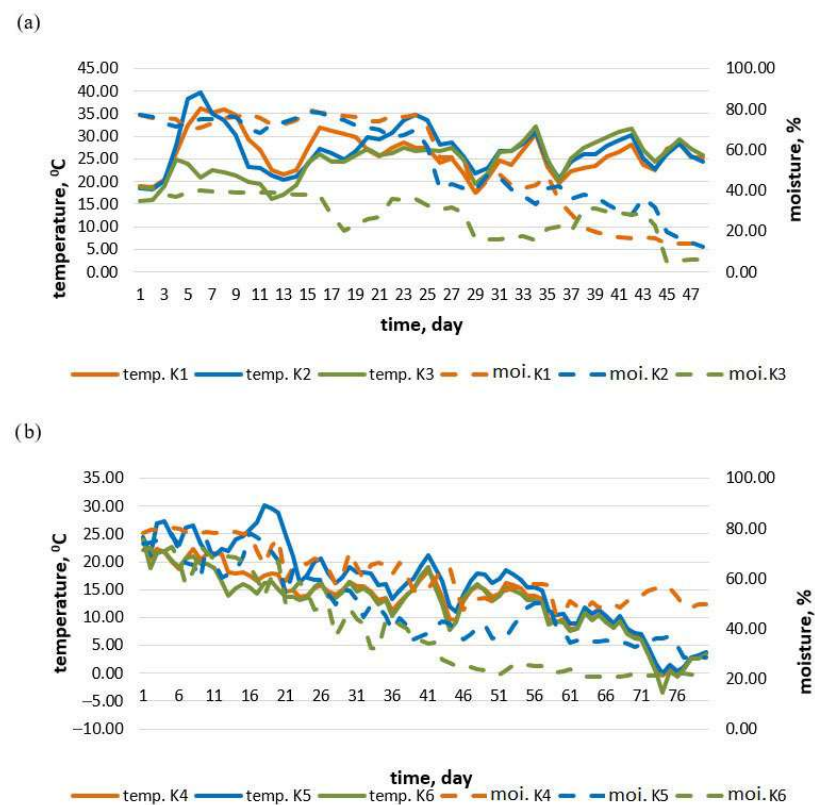


Figure 3. Changes in temperature and moisture content of the tested mixtures in the first (a) and second (b) research series.

The change in temperature of the mixtures was associated with the change in the temperature outside the dryer, as the indoor space was not additionally heated. The system was based on the so-called greenhouse effect, which caused the indoor temperature and the temperature of the mixtures to exceed the outdoor temperature, allowing for gradual water loss (Figure 2). In series 2, due to the lower external temperature, the process was slower, but it was possible to obtain the mixture parameters necessary for further use.

Series 1

Based on analyses of the moisture content of the mixtures in the first series of tests, the drying curves were plotted and the drying rate of each mixture was calculated. The results for series 1 of the tests are shown in Figure 4.

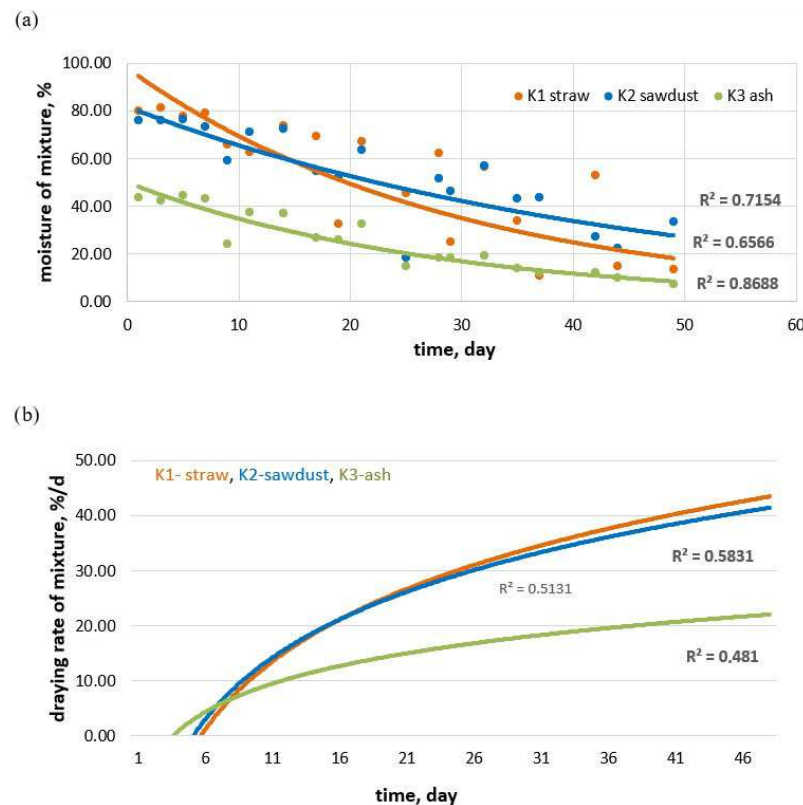


Figure 4. The drying process of the mixtures in the first series of research: (a) drying curves, (b) summed drying rate curves.

The initial hydration of the mixtures of sewage sludge with straw and sawdust was similar, at ca. 80%. Due to the nature of the additive, the lowest initial hydration was found for the mixture of sludge and ash—about 45%. The initial moisture contents of all three additives (straw, sawdust, and ash) were similar, at ca. 8%. The best results during drying were obtained for the mixture of sludge and ash; the ash was non-absorbent and did not adsorb water from the environment. The initial structure of the mixture was homogeneous; during drying and due to mixing, it changed into a highly porous one, resulting in a final hydration of less than 10%. Sawdust, like straw chaff, formed a homogeneous mixture when added to the sludge. These mixtures reduced the moisture content relatively slowly and, due to the more porous structure, the final effect was better for the K1 mixture, even though the initial hydration was a few percent higher. The rate of change in the moisture content of the mixtures over time was greatest at the beginning of the experiment (during the first several days), as can be seen in Figure 3b. As the moisture content of the mixtures was reduced, the drying rate gradually decreased. The significant difference for the K3 mixture compared to the K1 and K2 mixtures was due to the lower initial hydration and

the significantly higher organic content of the components for the K1 and K2 mixtures (sm. org. above 90%). The drying rate of the K1 mixture was slightly higher due to greater water loss after the 15th day of drying.

It was found that the temperature inside the tunnel was most influenced by insolation and outdoor temperature, to which the moisture content of the dried mixtures, especially the K1 and K2 mixtures, was also related. The moisture content of the mixtures was also satisfactorily related to the internal temperature, while its relationship with the internal moisture content and mixture temperature was small. Thus, it can be concluded that insolation and ambient temperature had the greatest influence on the drying of mixtures in the first series of tests, with the lowest correlation coefficient obtained for the K3 (ash) mixture (Table 3).

Table 3. Correlation of drying parameters in research series 1.

	Insolation	Outside Temp.	Inside Temp.	Inside Humidity	Temp. of Mixture
Outside temp.	0.4768				
Inside temp.	0.6044	0.9477			
Inside humidity	−0.7661	−0.4081	−0.3931		
Humidity of mixture	K1	−0.6777	−0.5732	0.1445	0.2961
	K2	−0.6984	−0.6039	0.0768	0.0916
	K3	−0.5005	−0.4681	0.0316	−0.4665

Series 2

Based on analyses of the moisture content of the mixtures in the second series of tests, the drying curves were plotted and the drying rate of each mixture was calculated. The results for series 2 of the tests are shown in Figure 5.

In the second series of drying sludge mixtures, the initial moisture contents of mixtures containing pine bark, wood chips, and walnut shells were similar, ranging from 70 to 80%. Walnut shells produced the best structure for drying sludge mixtures. The shells had very low absorbance, and the sewage sludge did not adhere to the surface of the shells, which made it possible to obtain a highly porous mixture; the shells were not visibly degraded during drying. Beech and alder woodchips also produced a porous mixture, but they had a greater ability to absorb moisture, so the drying curve results for the K5 mixture were worse than those for the K6 mixture. Pine bark had a much higher initial moisture content than the other components (ca. 50%), which influenced the high moisture content values of the mixture. The drying rates of the mixtures in the second series of tests differed quite considerably. The results turned out to be analogous to the drying effect: the lowest values were obtained for the mixture with bark and the highest values were obtained for the mixture with nut shells. The rate was highest at the beginning of the drying process and decreased with time.

It was found that the temperature inside the tunnel was most influenced by insolation and outdoor temperature, to which the moisture content of the dried mixtures was also related to a satisfactory degree. The moisture content of the mixtures was highly correlated with the air temperature inside the tunnel and the temperature of the mixtures, while it was very slightly correlated with the humidity inside the tunnel, with the highest correlation coefficient found for mix K4 (bark). Thus, it can be concluded that the process of drying the mixtures in the second series of tests was influenced by insolation, ambient temperature, and temperature of the dried material. The last factor may have been related to the lower ambient temperature at the time (Table 4).

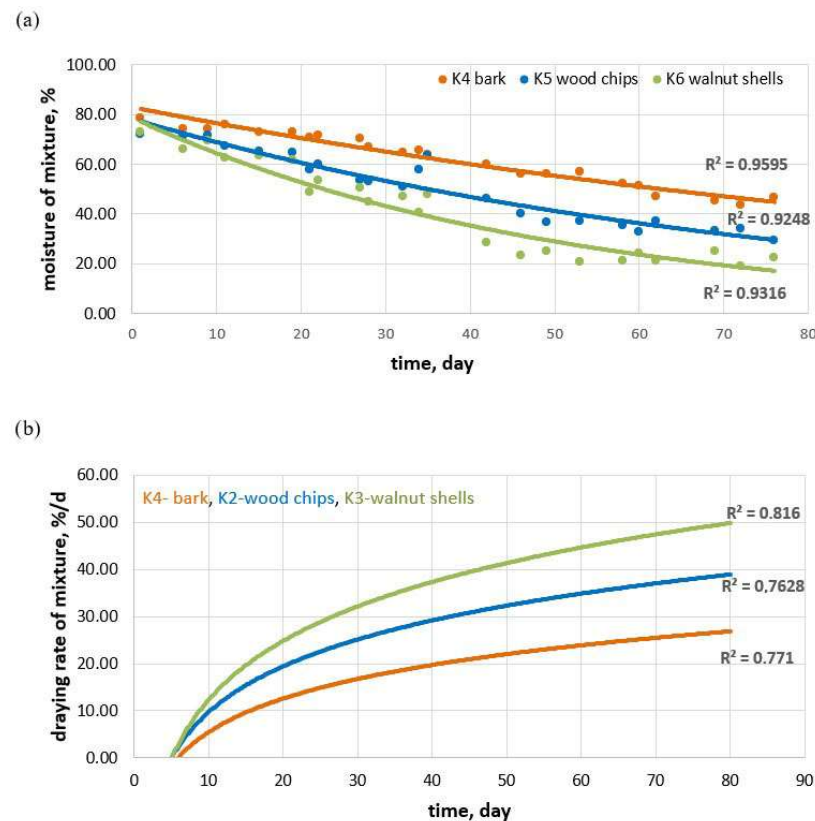


Figure 5. The drying process of the mixtures in the second series of research; (a) drying curves, (b) summed drying rate curves.

Table 4. Correlation of drying parameters in research series 2.

	Insolation	Outside Temp.	Inside Temp.	Inside Humidity	Temp. of Mixture
Outside temp.	0.6241				
Inside temp.	0.7537	0.9080			
Inside humidity	−0.5842	−0.2500	−0.1506		
Humidity of mixture	K4	0.4817	0.7385	−0.2351	0.7394
	K5	0.5585	0.7898	−0.2262	0.8277
	K6	0.5313	0.7749	−0.1774	0.7073

4. Discussion

The experiment was conducted to determine the effect of external parameters on the process of drying activated sludge mixtures with generally available components. The experiments were conducted in spring-summer (series 1) and fall (series 2). In series 1, the best effect was obtained for the K3 mixture (with the addition of ash), while slightly worse effects were obtained for the K1 and K2 mixtures (with the addition of straw chaff and sawdust, respectively). Here, we could see a clear effect of the initial hydration of the mixture and the organic matter content of the component, with the effect from the 15th day of drying the K1 mixture being better (the rate of drying of this mixture was also slightly higher). The drying speed of the K3 mixture was significantly lower, which is why it was considered less preferable than the other mixtures tested in this series. In addition, mixing this material was problematic as it caused dust to be stirred up. In series 2, despite similar initial hydration levels, the best result was obtained for mix K6 (with the addition

of walnut shells), slightly worse for mix K5 (with wood chips), and least favorable for mix K4 (with bark). This was due to the nature of the added components, their different absorbability, and initial moisture content, which made it possible to create mixtures with different porosities.

Table 5 shows the results of the analysis of the influence of external factors on the process of drying the tested mixtures in series 1 and 2, as measured by the changes in their moisture content. In both cases, the tunnel's internal temperature was shaped by insolation and external temperature. These factors had a strong or satisfactory effect on the moisture content of the mixtures, and therefore on the drying process. Hence, it follows that they were crucial for changing the moisture content of dried materials, which translated into the final outcome of the process. The drying effect was thus directly influenced by the ambient temperature, shaped by the degree of insolation. The season in which the process was carried out, due to differences in insolation, affected the time to achieve the desired drying effect.

Table 5. Impact of outside factors on the humidity of mixtures during the solar drying process.

<i>Factor</i>	Series 1			Series 2		
	K1	K2	K3	K4	K5	K6
<i>Insolation</i>	+++	+++	+++	++	++	++
<i>Outside temperature</i>	+++	+++	++	++	++	++
<i>Inside temperature</i>	++	++	++	+++	+++	+++
<i>Inside Humidity</i>	+	+	+	+	+	+
<i>Temperature of mixture</i>	+	+	++	+++	+++	+++

+ minor impact.; ++ satisfactory impact.; +++ major impact.

For both series of tests, it was found that an increase in insolation resulted in a decrease in humidity in the tunnel, but the drying rate of the tested mixtures was only slightly affected by the humidity of the immediate environment. The intensity of water evaporation from the surface of the mixture probably had a much greater effect on the moisture content.

A large difference between the results was found for the dependence of the moisture content and drying effect of the mixtures on the temperature of the material itself. It turned out that in series 1, the dependence of the mixture's moisture content on its temperature was small, and it was satisfactory only for the K3 mixture. By contrast, a large impact of mixture temperature on moisture content was found in series 2, which may have been related to the lower temperature, higher ambient humidity, and less evaporation of water from the surface of the mixtures.

Based on the above analyses, it was concluded that among the additives used, walnut shells proved to be the most favorable component due to the drying effect of the mixture, despite the lower outdoor temperature during the fall period. The drying rate of this mixture was comparable to that obtained for the straw mixture at a higher outdoor temperature, although the drying time to a moisture content of about 20% was half as long. Straw chaff proved to be the best component in the 1st round of drying; satisfactory results in the shortest time were obtained for the K1 mixture.

Mixtures with organic components such as straw, bark, wood chips, or sawdust can be used after drying for reclamation of post-mining areas, for example, or for agricultural purposes to improve soil quality. Similarly, a mixture with ash can be used, keeping in mind the much lower organic matter content of this mixture. The mixture with added nut shells can be used for agricultural purposes after fragmentation, while its use for energy generation purposes is beneficial, as are other mixtures with organic components. The

average calorific value of such biofuel is only about 30% less than the average calorific value of coal. A definite advantage is the ability to store mixtures after drying; their volume is reduced by several to more than a dozen times, which reduces storage costs.

In order to intensify drying and reduce the residence time in the dryer, the process can be supported thermally using waste heat, such as that recovered from wastewater fed to the treatment plant, thereby increasing the ambient temperature, which has the greatest impact on the drying effect.

5. Conclusions

The main cost in wastewater treatment plants is energy consumption. For this reason, it is necessary to adopt solutions related to energy optimization [52]. Sludge is a natural consequence of wastewater treatment processes. One method of preparing sludge for management and use is drying, including solar drying. Dried sewage sludge can be a valuable source of energy [53]. A more effective process can be achieved by co-drying with additives such as straw chaff, wood sawdust, ash, bark, wood chips, or walnut shells. Drying carried out in two series, at maximum insolation of 24.1 and 29.8 MJ·m⁻², respectively, allowed the evaluation of the influence of external parameters such as temperature, humidity, and insolation on the time, drying rate, and final drying effect. With an initial humidity of the mixtures of 41 to 79%, a final humidity of 6 to 49% was obtained, depending on the type of components and drying conditions. The drying rate was most influenced by the amount of solar energy, the associated outdoor (maximum 29 and 19 °C) and indoor (maximum 33 and 24 °C) air temperatures, and the temperature of the mixtures (maximum 30 °C), upon which the intensity of water evaporation depended.

The best additives in the drying process were straw chopped straw and walnut shells, for which the highest drying rates were obtained (maximum 50 to 60%/d). These materials created a beneficial mixture structure, enabling rapid loss of water contained in the sludge. The dried material can be used for agricultural purposes as a soil improver and for energy production. However, when using sewage sludge as a soil improver, its quality should be regularly monitored, as discussed by Nunes et al. [54], among others. Due to the cost of obtaining additional biomass, it is necessary to analyze not only the technical but also the economic aspects [55]. This was confirmed by the authors' earlier research [42,43]. Drying causes the mixture to reduce in volume, which means that it can be stored temporarily and is easier to transport, as confirmed by research [20]. Solar drying also provides additional sanitization of the material.

The proposed sewage sludge treatment and utilization method is part of circular economy waste management. This approach is in line with the principles of sustainable development. The sludge, which is often produced in large quantities, can be reused for environmental, agricultural, or energy purposes. This reduces the consumption of synthetic fertilizers and mineral fuels.

6. Patents

The results of the published research were included in patent application no. P.445919: Technology for using natural materials for solar drying of sewage sludge and production of sewage sludge-based briquettes.

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